When influencers feel like friends: Parasocial relationships and impulse buying tendency in live shopping

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Abstract

Purpose: This study examines susceptibility to impulse buying among Generation Z consumers in live shopping. It focuses on perceived value, serendipity and perceived scarcity, and assesses whether their associations with impulse buying tendency vary across parasocial relationships and self-esteem.

Design/methodology/approach: Survey data from 1,489 Portuguese Generation Z consumers were analysed using partial least squares SEM to estimate prediction-oriented associations among marketplace cues, parasocial relationships, self-esteem and impulse buying tendency.

Findings: Perceived value, serendipity and perceived scarcity were positively associated with impulse buying tendency. Parasocial relationships strengthened the associations involving perceived value and serendipity, but not perceived scarcity. Self-esteem weakened the associations between all three cues and impulse buying tendency. This pattern suggests that relational attachment and self-evaluation condition consumer susceptibility in live shopping.

Practical implications: Live shopping strategies should emphasise credible value communication, discovery-oriented presentation and transparent scarcity claims. Influencer relationships may increase responsiveness to value and discovery cues, making authenticity, disclosure and consumer autonomy important design considerations.

Social implications: The study positions live shopping as a social environment in which commercial persuasion is shaped by relational attachment and self-evaluation. This raises concerns for young consumers' autonomy and wellbeing when value, discovery and scarcity cues are reinforced by influencer intimacy. Ethical live commerce should support transparency, credible host conduct and reflective consumption.

Originality/value: The study advances impulse buying research by moving beyond an additive account of marketplace cues to an account of consumer susceptibility in live shopping. It shows that similar live shopping cues are associated with different levels of impulse buying tendency depending on consumers' parasocial attachment and self-esteem. Using Identity-Based Motivation as a conceptual lens, the article offers an identity-oriented interpretation of why marketplace cues may become more salient for some consumers than for others.

Keywords: impulse buying tendency; Identity-based motivation; parasocial relationships; self-esteem; live shopping; digital consumer behaviour; Generation Z

1. Introduction

Digital consumption is undergoing substantial transformation, with implications for consumer behaviour and commercial activity (Yi et al., 2023). Among emerging formats, live shopping has become one of the most immersive expressions of social commerce, combining entertainment, interpersonal interaction and real-time transactions within a single digital environment (Bao and Zhu, 2023; Sun et al., 2019). Unlike conventional electronic commerce interfaces, where consumers encounter static product displays, live shopping enables synchronous exchanges between hosts and audiences, creating conditions of immediacy, social presence and continuous stimulation (Chan et al., 2017; Li et al., 2022). These environments have been associated with spontaneous purchasing, especially among younger consumers (Feng et al., 2024; Pacheco et al., 2025).

Impulse buying is commonly described as a spontaneous purchase accompanied by affective arousal and reduced deliberation (Beatty and Ferrell, 1998; Redine et al., 2023). In digital commerce settings, this phenomenon has gained salience as platforms intensify immediacy, emotional stimulation and sustained exposure to promotional content (Yi et al., 2023). Prior research has linked impulsive buying to perceived value, hedonic gratification, fear of missing out, serendipitous discovery and scarcity cues (Bashar, Singh and Pathak, 2023; Ekayani et al., 2024; Prawira and Sihombing, 2021; Song et al., 2021). Much of this work is informed by stimulus-response reasoning, including the Stimulus-Organism-Response framework, where environmental cues are associated with internal states and consumer responses (Mehrabian and Russell, 1974).

This literature has generated considerable insight, but an important question remains. Consumers exposed to similar marketplace conditions do not report identical levels of impulse buying tendency. Research on parasocial relationships in influencer marketing and social commerce has expanded considerably, showing that consumers often form emotionally charged, one-sided connections with influencers that are associated with trust, engagement and purchase intention (Farivar et al., 2021; Jin and Ryu, 2020; Kim and Chan-Olmsted, 2022; Xiang et al., 2016). Research on self-esteem has also shown that consumers differ in their susceptibility to impulsive consumption (Dhandra, 2020; Zhang, 2023). Yet these relational and dispositional factors are often examined as predictors in their own right. Less is known about whether established marketplace cues are associated with impulse buying tendency differently across consumers who vary in parasocial attachment and self-evaluation.

The present study draws on Identity-Based Motivation as a sensitising framework for examining this variation. Identity-Based Motivation suggests that situational cues become more consequential when they connect with the self, shaping what feels relevant, appropriate or worth acting upon in a given setting (Oyserman, 2009; Oyserman, 2015). In consumer contexts, products, influencer recommendations and promotional cues may acquire salience through taste, aspiration, belonging and social positioning. Live shopping is therefore a useful setting for examining cue susceptibility, because product information is encountered within a social scene shaped by host visibility, audience participation and real-time interaction.

In this study, Identity-Based Motivation is used as a conceptual lens, not as a measured psychological mechanism. The survey does not measure self-relevance, identity congruence or perceived fit between products and the consumer's self-concept. The article therefore uses Identity-Based Motivation to organise an identity-oriented interpretation of conditional associations. This framing complements established explanations of impulse buying based on affective arousal, urgency, self-regulation and

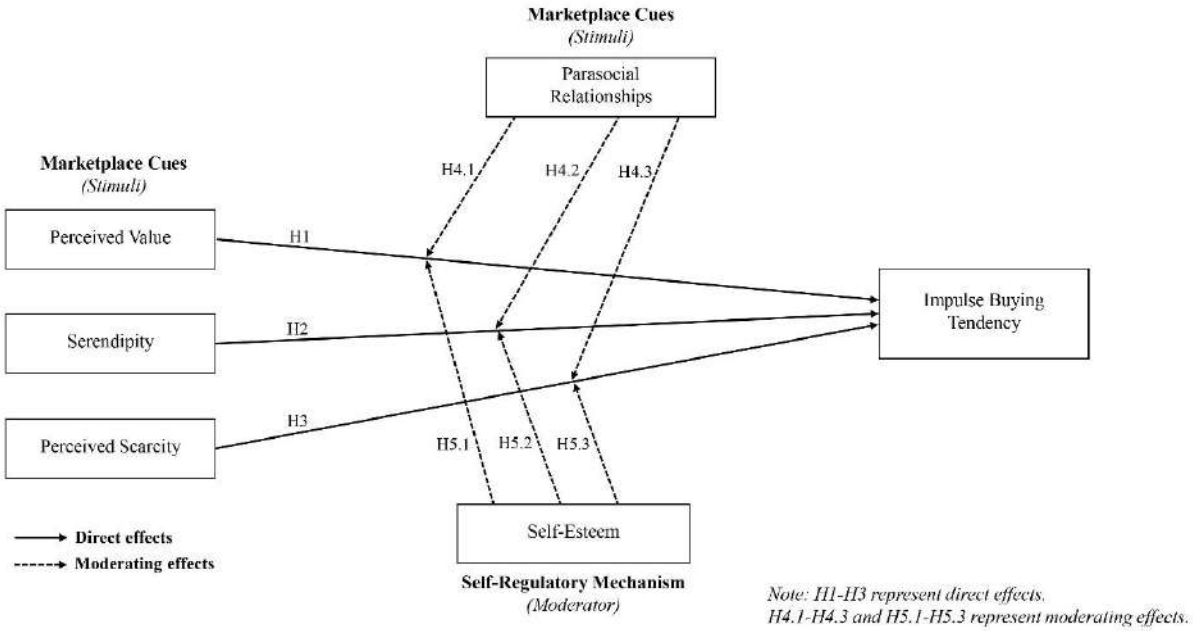
goal conflict (Higgins, 1997; Mehrabian and Russell, 1974; Verplanken and Sato, 2011). The study is not designed to choose among these theories. It asks whether cue-tendency associations vary across relational and dispositional conditions that prior theory links to social validation, external affirmation and the self.

The study's contribution lies in moving beyond an additive account of impulse buying antecedents. Perceived value, serendipity and perceived scarcity are well established in the literature. What remains less developed is how these marketplace cues operate in socially interactive live shopping environments where consumers differ in their attachment to hosts and in their reliance on external affirmation. We therefore examine perceived value, serendipity and perceived scarcity as marketplace signals, parasocial relationships as a form of relational attachment to the live shopping host, and self-esteem as a dispositional resource linked to self-evaluation. The model assesses whether associations between marketplace signals and impulse buying tendency vary across these consumer-level conditions.

Generation Z offers an especially appropriate context for examining these dynamics. As digital natives, Generation Z consumers are deeply embedded in socially interactive online environments characterised by continuous connectivity, influencer engagement and identity-oriented consumption practices (Malik et al., 2024). Prior research suggests that these characteristics increase susceptibility to impulsive buying, especially in environments that combine social interaction and immediacy (Yi et al., 2023). At the same time, live shopping remains less examined in European consumer contexts than in Asian markets, where the format is more mature. Portugal therefore offers a valuable setting for studying live shopping among digitally engaged young consumers in an evolving European market.

Accordingly, this study examines Portuguese Generation Z consumers who engage with live shopping. It assesses whether perceived value, serendipity and perceived scarcity are associated with impulse buying tendency, and whether these associations vary across parasocial relationships and self-esteem. In doing so, the article offers an account of live shopping susceptibility in a European consumer context. Figure 1 presents the conceptual model.

Figure 1. Conditional model of impulse buying tendency in live shopping



2. Revisiting Impulse Buying Tendency in the Context of Live Shopping

Impulse buying has long been described as a spontaneous, affectively charged purchase accompanied by reduced deliberation (Chan et al., 2017; Rook and Hoch, 1985). Early accounts emphasised sudden urges triggered by marketplace stimuli and constrained by limited cognitive control (Beatty and Ferrell, 1998). Subsequent research has broadened this view by incorporating situational factors, such as promotions and scarcity cues, and consumer characteristics, such as personality traits, buying impulsiveness and self-regulation (Kimiagari and Malafe, 2021; Redine et al., 2023).

In the present study, the dependent construct is impulse buying tendency. This distinction matters because the Verplanken and Herabadi (2001) scale captures a general predisposition towards spontaneous and less deliberative purchasing, not an observed purchase during a particular live shopping episode. The study therefore examines whether perceptions formed in live shopping are associated with variation in this tendency among consumers who engage with that format. It does not claim to observe actual purchase conduct in real time.

This focus is well suited to live shopping. The format combines audiovisual demonstration, influencer interaction, audience visibility and time-sensitive offers, creating conditions in which marketplace cues can become salient in rapid evaluative moments (Bao and Zhu, 2023; Li et al., 2022; Sun et al., 2019). Perceived value, serendipity and scarcity capture three distinct types of cue in this setting: evaluative appraisal, unexpected discovery and perceived constraint. Prior impulse buying research has examined these cues mainly as antecedents. Less attention has been given to whether their associations with impulse buying tendency vary across relational and dispositional conditions.

This study uses Identity-Based Motivation to organise that conditional account. The theory is relevant here because it treats the significance of situational cues as dependent on how they connect with the self in a given setting (Oyserman, 2009; Oyserman, 2015). In live shopping, products and promotional claims are encountered within a social scene shaped by influencer presence and audience participation. Marketplace cues may therefore acquire salience through taste, aspiration, belonging and social positioning. Since the survey did not measure self-relevance or identity congruence, the study uses Identity-Based Motivation as a conceptual lens for interpreting cue susceptibility, not as a measured process.

Parasocial relationships and self-esteem are positioned as boundary conditions in this account. Parasocial relationships capture the viewer's relational attachment to the live shopping host and may strengthen the association between marketplace cues and impulse buying tendency by increasing trust, familiarity and perceived social validation (Farivar et al., 2021; Hoffner and Bond, 2022). Self-esteem captures a dispositional resource associated with the need for external affirmation and may weaken these associations by allowing greater psychological distance from value claims, unexpected discoveries and scarcity pressures (Dhandra, 2020; Zhang, 2023).

2.1 The Role of Perceived Value in Impulse Buying Tendency

Perceived value is a long-standing concept in consumer behaviour research and is commonly defined as consumers' subjective appraisal of the balance between perceived benefits and associated costs (Dodds et al., 1991; Qiao, Yin and Xing, 2022; Walsh et al., 2014). Contemporary research treats perceived value as multidimensional, encompassing

functional, emotional and social dimensions that shape consumer responses (Leroi-Werelds, 2019; Sweeney and Soutar, 2001). Functional value concerns product performance and utility, emotional value concerns affective responses associated with consumption, and social value concerns symbolic expression, social standing and belonging (Zuliarni et al., 2023).

In live shopping environments, value cues are made more vivid through audiovisual product demonstration, host endorsement and real-time interaction (Li et al., 2022; Sun et al., 2019). These features can reduce uncertainty and accelerate product evaluation, while placing value assessments within a socially interactive setting. Value is therefore not assessed only as price utility or product quality. It is encountered in a setting where host credibility, audience participation and social approval may make a product appear more attractive and easier to justify.

From an Identity-Based Motivation lens, value cues may gain salience when they resonate with taste, competence, belonging or social standing (Escalas and Bettman, 2005; Oyserman, 2015). In live shopping, perceived value is therefore likely to be associated with impulse buying tendency because product evaluation occurs in a rapid, socially visible and host-centred setting.

H1: Perceived value is positively associated with impulse buying tendency among Generation Z consumers who engage with live shopping.

2.2 Serendipity as Unexpected Discovery

Serendipity refers to the experience of encountering unexpected yet valuable outcomes (Liu et al., 2022; Yaqub, 2018). In consumer behaviour research, it has been examined as unplanned product discovery associated with curiosity, excitement and enjoyment (Kim et al., 2021). Such encounters can interrupt routine evaluation and increase openness to spontaneous purchasing, especially when discovery is affectively engaging (Prawira and Sihombing, 2021; Yi et al., 2023).

In digital commerce, serendipity is often shaped by platform design. Algorithmic recommendations, personalised feeds and interface features can expose consumers to products they did not seek but may still find attractive (Redine et al., 2023). In live shopping, this experience becomes more social. Discovery unfolds through the host's presentation, audience comments, visible reactions and the rhythm of the stream (Bao and Zhu, 2023). An unexpected product can therefore appear not only novel, but also timely, socially endorsed and worth considering in the moment.

Prior research has mainly examined serendipity as an affective source of impulsive consumption, emphasising enjoyment, curiosity and arousal (Kim et al., 2021; Yi et al., 2023). The present study adds a conditional perspective. From an Identity-Based Motivation lens, unexpected discoveries may gain salience when they resonate with exploration, self-expression or participation in a shared shopping experience (Escalas and Bettman, 2005; Oyserman, 2015). In live shopping, serendipity is therefore expected to be positively associated with impulse buying tendency because discovery occurs in a socially charged setting where novelty and participation are intertwined.

H2: Serendipity is positively associated with impulse buying tendency among Generation Z consumers who engage with live shopping.

2.3 Scarcity Cues and the Psychology of Urgency in Impulsive Buying Tendency

Perceived scarcity refers to the perception that a product is limited in quantity or availability, creating urgency in consumer evaluation (Biraglia et al., 2021). Scarcity cues reduce the time available for comparison and increase the perceived cost of inaction, which can encourage faster purchase decisions (Shi et al., 2020). These effects are often explained through Psychological Reactance Theory, which suggests that perceived restrictions on choice can create motivation to restore a sense of freedom (Brehm, 1989).

Scarcity also carries social and symbolic value. Limited availability can signal exclusivity, distinction and status, making ownership appear less routine and more socially desirable (Gupta and Gentry, 2016; Musadik, 2021; Omar et al., 2021). Prior research further suggests that scarcity becomes more salient under perceived competition and social presence (Song et al., 2021).

In live shopping, scarcity is displayed through countdown timers, limited stock notifications and visible purchase activity (Feng et al., 2024; Pacheco et al., 2026). These cues make urgency public and synchronous. Consumers can see that others are watching, reacting and buying, which may intensify the perception that delay involves loss of access to a valued opportunity.

Although scarcity effects are well established, live shopping gives scarcity a socially visible form. From an Identity-Based Motivation lens, scarcity cues may gain salience when limited availability is associated with access, distinction or inclusion (Oyserman, 2015). Perceived scarcity is therefore expected to be positively associated with impulse buying tendency among consumers who engage with live shopping.

H3: Perceived scarcity is positively associated with impulse buying tendency among Generation Z consumers who engage with live shopping.

2.4 Building One-Sided Bonds: Parasocial Relationships in Live Shopping Contexts

Parasocial relationships are an important relational feature of live shopping environments (Hoffner and Bond, 2022; Hu et al., 2020; Mardon et al., 2023). These emotionally charged, one-sided connections between viewers and influencers can make commercial encounters feel more familiar, socially embedded and personally salient (Kim and Chan-Olmsted, 2022). Compared with conventional advertising, live shopping reduces perceived distance between seller and audience through continuous interaction, host visibility and real-time communication (Farivar et al., 2021). Consumers may therefore experience a sense of interpersonal connection, even in the absence of reciprocal relationships (Hoffner and Bond, 2022; Sheng et al., 2025; Tukachinsky, 2011).

Existing research has often examined parasocial relationships as predictors of engagement, trust and purchase intention (Chung and Cho, 2017; Djafarova and Bowes, 2021; Lee and Watkins, 2016). The present study takes a more conditional view. It examines whether parasocial relationships qualify the associations between marketplace cues and impulse buying tendency. This positioning is consistent with live shopping as a socially interactive setting in which product cues are presented through a host whom viewers may perceive as familiar, credible and socially affirming.

Identity-Based Motivation provides a useful lens for this argument because it suggests that situational cues may become more salient when they connect with the self in a given setting (Oyserman, 2015). In live shopping, parasocial attachment may increase the salience of value claims, product discoveries and scarcity messages by embedding them

in a relational context. The argument is not that parasocial relationships prove an identity process. It is that they may condition the strength of cue tendency associations by making marketplace cues feel more familiar, trusted and personally salient.

2.4.1 Parasocial Relationships as Moderating Mechanisms Between Perceived Value and Impulse Buying Tendency

Within live shopping environments, parasocial relationships may shape how consumers respond to value-related marketplace cues. When viewers feel attached to a live shopping host, product information may be received with greater familiarity and trust (Hoffner and Bond, 2022). Value claims are then less likely to appear as detached commercial information and more likely to be encountered as recommendations embedded in an ongoing relational experience (Farivar et al., 2021).

This relational context may strengthen the association between perceived value and impulse buying tendency. Value-related cues communicated by a trusted host can make product evaluation faster and more favourable, especially when the host is perceived as credible and similar to the audience (Hoffner and Bond, 2022; Hu et al., 2020). In such settings, perceived value may provide not only economic justification but also social reassurance, making spontaneous purchasing feel more acceptable in the moment (Ekayani et al., 2024; Gunawan and Iskandar, 2020; Mardon et al., 2023).

The moderating role of parasocial relationships is therefore framed as relational qualification. Stronger parasocial bonds may make value cues more personally salient, strengthening the association between perceived value and impulse buying tendency.

H4.1: Parasocial relationships positively moderate the association between perceived value and impulse buying tendency.

2.4.2 Parasocial Relationships and Serendipitous Discovery: Turning Surprise into Action

Serendipitous discovery is a highly engaging aspect of live shopping because unexpected yet desirable encounters can generate curiosity, excitement and attention (Kim et al., 2021; Prawira and Sihombing, 2021; Yi et al., 2023). In live shopping, such discoveries unfold through host presentation, audience reactions and the rhythm of the stream. The same unexpected product may therefore be experienced differently depending on the viewer's relational attachment to the host.

Parasocial relationships may strengthen the association between serendipity and impulse buying tendency by placing discovery within a trusted relational setting. When viewers feel connected to the host, unexpected product encounters may appear less random and more socially endorsed (Djafarova and Bowes, 2021; Hoffner and Bond, 2022). The host's enthusiasm and credibility can make discovery feel timely, attractive and worth considering.

For consumers with stronger parasocial relationships, the affective appeal of serendipity may therefore become linked with trust, familiarity and social participation (Jin and Ryu, 2020; Liu et al., 2022). Unexpected encounters are then more likely to be associated with impulse buying tendency. When parasocial attachment is weaker, serendipity may still capture attention, but its association with impulse buying tendency may be less pronounced.

H4.2: Parasocial relationships positively moderate the association between serendipity and impulse buying tendency.

2.4.3 Parasocial Relationships and the Perception of Scarcity: Trust Under Pressure

Scarcity cues are commonly associated with urgency in live shopping, as countdown timers, limited stock messages and exclusive offers compress decision time and heighten perceived competition (Akram et al., 2023; Feng et al., 2024; Yi et al., 2023). Yet exposure to scarcity does not necessarily relate to impulse buying tendency in the same way for all consumers. Some viewers may respond to scarcity as useful information about availability, whereas others may experience it as social pressure or a risk of missing out.

Parasocial relationships may qualify this association by embedding scarcity cues within a trusted host relationship. When viewers feel attached to an influencer, scarcity messages may be received less as generic promotional pressure and more as guidance from a familiar source (Farivar et al., 2021; Hu et al., 2020). This relational setting may reduce scepticism and increase responsiveness to scarcity appeals.

Parasocial attachment may therefore strengthen the association between perceived scarcity and impulse buying tendency. Scarcity cues communicated by a trusted host can become socially salient through interaction, audience visibility and perceived collective participation (Kim and Chan-Olmsted, 2022; Ming et al., 2021; Xiang et al., 2016). In this setting, limited availability may be associated not only with urgency, but also with access to valued experiences, distinction or inclusion (Biraglia et al., 2021; Chen et al., 2023).

At the same time, scarcity may depend less on host attachment than value or discovery, because urgency and time pressure can operate even when relational attachment is weak. This possibility makes the scarcity interaction theoretically informative and worth examining in the live shopping context.

H4.3: Parasocial relationships positively moderate the association between perceived scarcity and impulse buying tendency.

2.5 The Role of Self-Esteem in Shaping Consumer Vulnerability

Self-esteem has been widely examined in consumer behaviour research as a factor associated with impulsive buying tendencies (Dhandra, 2020; Reitz et al., 2020; Zhang, 2023). Much of this work treats self-esteem as an individual difference linked to susceptibility to compensatory consumption, affective purchasing and reduced self-control. Less attention has been given to how self-esteem may qualify the association between marketplace cues and impulse buying tendency in socially interactive shopping environments.

This issue is especially relevant in live shopping. The format combines host interaction, audience visibility, time pressure and emotional stimulation, creating a commercial setting in which consumers encounter value claims, unexpected discoveries and scarcity cues in rapid succession (Parameswari et al., 2024). Under these conditions, consumers may differ in the psychological distance they maintain from marketplace cues. Self-esteem is relevant because it captures a broad evaluation of personal worth and is associated with dependence on external affirmation, sensitivity to social comparison and

confidence in one's own judgement (Dhandra, 2020; Reitz et al., 2020; Zhang and Hawk, 2022).

From an Identity-Based Motivation lens, self-esteem can be understood as a dispositional resource that shapes cue susceptibility. Consumers with higher self-esteem may be less reliant on external validation and more able to maintain evaluative distance from value claims, discovery appeals and scarcity pressure. Consumers with lower self-esteem may be more responsive to cues that appear to offer affirmation, gratification or protection from exclusion (Challet-Bouju et al., 2020; Zhang, 2023). The argument is therefore not that self-esteem determines impulse buying tendency by itself. It is that self-esteem may condition the strength of the association between marketplace cues and impulse buying tendency.

This role differs from that of parasocial relationships. Parasocial relationships capture relational attachment to the host and may make marketplace cues more familiar, trusted and socially affirmed. Self-esteem captures a dispositional resource associated with external affirmation and psychological distance. The two constructs therefore qualify cue susceptibility through different pathways: parasocial relationships through relational attachment, and self-esteem through self-evaluation.

2.5.1 Self-Esteem as a Psychological Lens on the Relationship Between Perceived Value and Impulse Buying Tendency

Perceived value can make spontaneous purchasing feel more acceptable by providing a justification for immediate purchase. In live shopping, this justification may be reinforced through product demonstration, host endorsement and audience response. Yet consumers may differ in how strongly value claims are associated with impulse buying tendency.

Self-esteem is likely to qualify this association. Consumers with lower self-esteem may be more responsive to value-related cues when those cues appear to offer self-enhancement, social approval or reassurance about the purchase (Walsh et al., 2014; Wiener et al., 2022; Zhang and Hawk, 2022). By contrast, consumers with higher self-esteem may maintain greater psychological distance from persuasive value claims and may be less dependent on consumption as a source of affirmation (Challet-Bouju et al., 2020; Dhandra, 2020).

In live shopping, where value claims are presented in a socially engaging format, self-esteem may therefore weaken the association between perceived value and impulse buying tendency.

H5.1: Self-esteem negatively moderates the association between perceived value and impulse buying tendency.

2.5.2 When Surprise Meets the Self: Self-Esteem and Serendipitous Impulse Buying Tendency

Serendipitous encounters introduce unplanned and emotionally engaging consumption opportunities, often characterised by novelty, curiosity and affective stimulation (Busch and Grimes, 2023; Kim et al., 2021). In live shopping, these encounters are embedded in host presentation, audience reaction and social participation, which can make unexpected discovery especially engaging (Hoffner and Bond, 2022; Hu et al., 2020).

Self-esteem may qualify the association between serendipity and impulse buying tendency. Consumers with higher self-esteem may be better able to experience surprise and enjoyment without treating the discovered product as something that must be purchased. They may evaluate whether the product fits their preferences and goals before acting on the appeal of discovery (Dhandra, 2020; Oyserman, 2015). Consumers with lower self-esteem may be more responsive to the affective intensity of discovery, especially when the encounter appears to offer self-enhancement, social affirmation or participation in a shared moment (Zhang, 2023; Zhang and Hawk, 2022).

In this sense, self-esteem may weaken the association between serendipity and impulse buying tendency by giving consumers greater distance from the emotional pull of unexpected discovery.

H5.2: Self-esteem negatively moderates the association between serendipity and impulse buying tendency.

2.5.3 Scarcity Perceptions and the Buffering Role of Self-Esteem

Scarcity cues introduce urgency by signalling limited availability and increasing the perceived cost of delay (Gupta and Gentry, 2016; Shi et al., 2020). In live shopping, these cues are made visible through countdown timers, limited stock messages, audience activity and perceived competition (Song et al., 2021). Scarcity can therefore make purchase opportunities appear time-sensitive and socially consequential.

Self-esteem may qualify how strongly scarcity is associated with impulse buying tendency. Consumers with higher self-esteem may be less dependent on possession, social approval or group inclusion as sources of self-worth, making scarcity cues less likely to generate strong susceptibility (Dhandra, 2020; Feng et al., 2024). Consumers with lower self-esteem may be more responsive to the social implications of limited availability, especially when scarcity is experienced as potential exclusion from a valued consumption opportunity (Goel et al., 2022; Rosenberg et al., 1995).

Accordingly, self-esteem may function as a buffer against scarcity pressure in live shopping, weakening the association between perceived scarcity and impulse buying tendency.

H5.3: Self-esteem negatively moderates the association between perceived scarcity and impulse buying tendency.

3. Materials and Methods

This study employed a quantitative research design to examine how parasocial relationships and self-esteem shape impulsive buying tendencies within live shopping environments. The analysis focused on Generation Z consumers who regularly participate in live shopping activities and engage with influencer-led content.

A purposive sampling approach was adopted to recruit individuals actively involved in live shopping environments. Participants were recruited through collaborations with eight live shopping hosts operating primarily on Instagram and TikTok within Portuguese digital commerce settings. Hosts were selected based on audience size, posting frequency, and regular involvement in live shopping activities targeting younger consumers. Participation was voluntary and occurred after participants were exposed to recruitment

links shared during live sessions. Consequently, individuals with stronger interest or engagement in live shopping may have been more likely to participate.

Data collection was conducted through Qualtrics between 1 December 2024 and 28 February 2025. Participation was voluntary, and responses remained anonymous throughout the study. To encourage participation while preserving anonymity, respondents were offered the opportunity to enter a separate giveaway for electronic products through an independent form that was not connected to survey responses.

Portugal provides a relevant empirical context for examining live shopping behaviour within European digital commerce environments. Existing research on live shopping has predominantly focused on Asian markets, whereas evidence from European contexts remains comparatively limited. Online shopping and influencer-led commerce have expanded considerably among younger Portuguese consumers in recent years (ANACOM, 2024), making this context especially appropriate for examining socially interactive digital consumption practices.

A screening procedure ensured that respondents regularly engaged with live shopping content and influencer-led purchasing environments. Participants who did not meet the established criteria were excluded from the analysis. Anonymity was emphasised throughout the data collection process to reduce potential social desirability effects associated with impulsive consumption behaviour. A pre-test involving fifteen participants was conducted to assess questionnaire clarity, response flow, and completion time, resulting in minor wording adjustments. All procedures complied with the ethical standards of the European Research Ethics Framework and received approval from an accredited institutional ethics committee.

The questionnaire comprised two sections. The first collected sociodemographic information, including gender, age, education, occupation, and household income. The second measured the constructs included in the conceptual model using previously validated scales: impulse buying tendency (Verplanken & Herabadi, 2001), perceived value (Walsh et al., 2014), serendipity (adapted from McCay-Peet & Toms, 2011), perceived scarcity (adapted from Wu et al., 2011), self-esteem (Rosenberg et al., 1995), and parasocial relationships with live shopping influencers (Aw et al., 2022). All items were translated, reviewed, and pre-tested to ensure semantic equivalence. Details regarding the measurement instruments are presented in Appendix A.

The study examines impulse buying tendency within live shopping environments instead of specific impulsive purchase episodes. Consistent with prior research, impulse buying tendency reflects a dispositional inclination towards spontaneous and less deliberative purchasing responses when consumers encounter marketplace stimuli (Verplanken & Herabadi, 2001). In the present context, the construct captures consumers' susceptibility to impulsive purchasing responses associated with live shopping environments, not the occurrence of specific purchase events.

Sample size adequacy was assessed following the recommendations of Soper (2025) and Westland (2010) for structural equation modelling. The sample-size calculation considered six latent constructs, thirty-seven observed indicators, an expected effect size of 0.12, statistical power of 0.80, and a significance threshold of 0.05. The minimum recommended sample size was 1,179 respondents. The final validated sample comprised 1,489 participants, exceeding the recommended threshold and providing sufficient statistical power for the proposed analyses. A post-hoc sensitivity analysis further indicated that the final sample size was adequate to detect small interaction effects at 80% power and $\alpha = 0.05$. Table 1 presents the demographic characteristics of the sample.

Table 1. Sample profile

	Number	%
Sex		
Male	606	40.7
Female	883	59.3
Total	1,489	100
Age		
18 to 29	1,489	100
Total	1,489	100
Marital status		
Single	576	38.7
Civil union	226	15.2
Married	598	40.1
Divorced	89	6
Total	1,489	100
Education level		
High school or lower	881	59.2
Higher education	608	40.8
Total	1,489	100
Occupation		
Employed	887	59.6
Students	493	33.1
Unemployed	109	7.3
Total	1,489	100
Household income		
Able to live comfortably and save	335	22.5
Able to live comfortably	691	46.4
Covers only essential expenses	234	15.7
Income is insufficient but manageable with difficulty	188	12.6
Unable to meet basic needs	41	2.8
Total	1,489	100
Most frequently used platforms for live shopping		
Facebook	599	40.2
Instagram	435	29.2
TikTok	281	18.9
YouTube	174	11.7
Total	1,489	100
Main product categories in live shopping		
Fashion	634	42.6
Beauty	198	13.3
Technology	237	15.9
Decor	137	9.2
Fitness	116	7.8
Children's Products	167	11.2
Total	1,489	100

The analysis was conducted using SmartPLS 4 (Ringle et al., 2024). Partial least squares structural equation modelling was selected because the study estimates a prediction-oriented model with several interaction terms linking marketplace cues, parasocial relationships, self-esteem and impulse buying tendency. This approach is appropriate for examining conditional associations in a complex model, while the interpretation of the findings remains aligned with the cross-sectional survey design. The analysis therefore focuses on predictive associations among constructs, not on covariance-based theory confirmation or causal ordering (Hayes, 2022).

Before estimating the model paths, reliability, convergent validity and discriminant validity were assessed in line with established recommendations for reflective measurement models (Hair et al., 2019). The dependent variable was impulse buying tendency. The Verplanken and Herabadi (2001) scale captures a dispositional inclination towards spontaneous and less deliberative buying. Respondents answered the items with recent live shopping experiences in mind, which anchors the measure in the empirical setting while preserving its status as a tendency measure.

Full collinearity VIF values were examined as a diagnostic for multicollinearity and possible common method concerns. These values are reported as an auxiliary diagnostic,

not as a test that rules out common method variance. The study used a single-source survey design, and the treatment of possible common method effects is therefore addressed further in the limitations section.

4. Results

4.1 Measurement Model Evaluation

The coefficient of determination (R^2) was used to assess the explanatory power of the endogenous constructs. The model explained 14% of the variance in self-esteem (R^2 = 0.14), 38% of the variance in parasocial relationships (R^2 = 0.38), and 67% of the variance in impulse buying tendency (R^2 = 0.67).

Measurement reliability and validity were assessed prior to structural model estimation. Table 2 presents Cronbach's alpha, composite reliability, and average variance extracted (AVE) values together with convergent and discriminant validity assessments. All reflective loadings exceeded the recommended threshold of 0.50, supporting indicator reliability. Cronbach's alpha values ranged from 0.88 to 0.93, while composite reliability values ranged from 0.88 to 0.94, exceeding the recommended threshold of 0.70 (Hair et al., 2019). All AVE values exceeded 0.50, supporting convergent validity (Ringle et al., 2024).

Table 2. Reliability and Discriminant Validity

	CA	CR	AVE	Construct correlations*					
				1	2	3	4	5	6
1. Impulse Buying Tendency	0.91	0.94	0.65	**0.800**	[0.61]	[0.66]	[0.58]	[0.63]	[0.49]
2. Serendipity	0.93	0.88	0.70	0.54	**0.819**	[0.59]	[0.51]	[0.57]	[0.41]
3. Perceived Value	0.88	0.93	0.66	0.61	0.56	**0.831**	[0.62]	[0.68]	[0.46]
4. Perceived Scarcity	0.89	0.88	0.66	0.49	0.45	0.53	**0.812**	[0.55]	[0.38]
5. Parasocial Relationships	0.90	0.92	0.66	0.58	0.52	0.64	0.50	**0.825**	[0.44]
6. Self-Esteem	0.91	0.92	0.63	-0.42	-0.35	-0.39	-0.31	-0.37	**0.794**

Note: Values in bold represent the square root of the AVE. HTMT ratios are presented in brackets, while Fornell-Larcker correlations are reported below the diagonal.

Discriminant validity was assessed using the Fornell-Larcker criterion (Fornell & Larcker, 1981) and the Heterotrait-Monotrait (HTMT) ratio (Henseler et al., 2015). In all cases, the square root of the AVE exceeded the corresponding inter-construct correlations, supporting discriminant validity. All HTMT ratios remained below the recommended threshold of 0.90.

4.2 Structural Model Evaluation

The structural model revealed statistically significant associations among most constructs included in the proposed framework. Model fit was assessed using the Standardised Root Mean Square Residual, with SRMR = 0.034, below the recommended threshold of 0.08 (Hair et al., 2019). Variance inflation factors were examined as a diagnostic for multicollinearity. As shown in Table 3, all VIF values remained below the recommended threshold of 3.3 (Kock, 2015), indicating that multicollinearity was not a major concern for model estimation. As noted in the Methods and Limitations sections, these values should not be interpreted as ruling out common method variance.

Table 3. Variance Inflation Factors (VIFs)

Predictor Construct	VIF
Perceived Value	2.14
Serendipity	2.08
Perceived Scarcity	1.87
Parasocial Relationships	2.31
Self-Esteem	1.76
Perceived Value × Parasocial Relationships	2.42
Serendipity × Parasocial Relationships	2.37
Perceived Scarcity × Parasocial Relationships	2.11
Perceived Value × Self-Esteem	2.05
Serendipity × Self-Esteem	1.98
Perceived Scarcity × Self-Esteem	1.92

Eight of the nine proposed associations received empirical support. Tables 4 and 5 present the structural model results.

Table 4. Direct Effects

Path	B	t-value	p-value	LLCI (5%)	ULCI (95%)	f^2	Empirical support
H1: Perceived Value -> Impulse Buying Tendency	0.24	4.85	< 0.001	0.16	0.32	0.10	Supported
H2: Serendipity -> Impulse Buying Tendency	0.29	5.72	< 0.001	0.21	0.37	0.13	Supported
H3: Perceived Scarcity -> Impulse Buying Tendency	0.21	4.96	< 0.001	0.14	0.28	0.08	Supported

Note. LLCI = lower-level confidence interval; ULCI = upper-level confidence interval. Confidence intervals are based on one-tailed bootstrapping using the 5% and 95% percentile bounds.

Table 5. Moderating Effects

Path	*B*	*t-value*	*p-value*	*LLCI (5%)*	*ULCI (95%)*	f^2	Empirical support
H4.1: Parasocial Relationships × Perceived Value -> Impulse Buying Tendency	0.11	2.74	0.006	0.04	0.18	0.024	Supported
H4.2: Parasocial Relationships × Serendipity -> Impulse Buying Tendency	0.09	2.31	0.021	0.03	0.15	0.019	Supported
H4.3: Parasocial Relationships × Perceived Scarcity -> Impulse Buying Tendency	0.04	1.287	0.198	-0.01	0.09	0.003	Not supported
H5.1: Self-Esteem × Perceived Value -> Impulse Buying Tendency	-0.10	2.12	0.034	-0.18	-0.02	0.021	Supported
H5.2: Self-Esteem × Serendipity -> Impulse Buying Tendency	-0.08	2.02	0.043	-0.15	-0.02	0.015	Supported
H5.3: Self-Esteem × Perceived Scarcity -> Impulse Buying Tendency	-0.07	2.00	0.046	-0.13	-0.01	0.012	Supported

The interaction terms show modest incremental effect sizes, consistent with their theoretical role as boundary conditions in a large survey model. Parasocial relationships qualify the associations involving perceived value and serendipity, while self-esteem qualifies the associations involving perceived value, serendipity and perceived scarcity. The results therefore indicate that live shopping cues are not associated with impulse buying tendency uniformly across consumers.

Perceived value was positively associated with impulse buying tendency, with $\beta = 0.24$, $t = 4.85$ and $p < 0.001$. This result is consistent with prior work showing that consumers' evaluations of product utility, quality and value can increase susceptibility to spontaneous

purchasing (Walsh et al., 2014; Ekayani et al., 2024; Zhang, 2023). In live shopping, value is communicated through product demonstration, host endorsement and real-time interaction. The finding suggests that value perceptions among Generation Z live shopping consumers may involve more than price utility. They may also provide a justification that makes spontaneous purchasing feel acceptable in the moment.

Serendipity was also positively associated with impulse buying tendency, with $\beta = 0.29$, $t = 5.72$ and $p < 0.001$. This result aligns with research showing that unexpected product discovery can generate curiosity, enjoyment and affective arousal (Kim et al., 2021; Prawira and Sihombing, 2021; Yi et al., 2023). In live shopping, discovery is rarely a private browsing event. It unfolds within a social stream where the host's presentation, audience reactions and platform flow can make an unexpected product appear timely, attractive and personally relevant. This helps explain why serendipity displays the strongest main association in the model.

Perceived scarcity was positively associated with impulse buying tendency, with $\beta = 0.21$, $t = 4.96$ and $p < 0.001$. This result is in line with studies showing that scarcity cues intensify urgency, perceived competition and fear of missing out (Gupta and Gentry, 2016; Shi et al., 2020; Song et al., 2021; Feng et al., 2024). In live shopping, scarcity is visible through countdowns, limited stock messages and real-time purchase signals. The finding implies that scarcity may be salient for Generation Z consumers because limited availability is displayed socially and synchronously, making urgency part of the shared shopping experience.

The moderation results involving parasocial relationships add a relational qualification to this pattern. Parasocial relationships strengthened the association between perceived value and impulse buying tendency, with $\beta = 0.11$, $t = 2.74$ and $p = 0.006$, and also strengthened the association between serendipity and impulse buying tendency, with $\beta = 0.09$, $t = 2.31$ and $p = 0.021$. These findings are consistent with research showing that parasocial relationships with influencers can increase trust, perceived closeness and receptiveness to recommendations (Xiang et al., 2016; Farivar et al., 2021; Jin and Ryu, 2020; Kim and Chan-Olmsted, 2022; Mardon et al., 2023). In live shopping, value claims and unexpected discoveries may therefore become more closely associated with impulse buying tendency when they are communicated within a relationship that viewers experience as familiar, credible and socially affirming.

The interaction between parasocial relationships and scarcity was not statistically significant, with $\beta = 0.04$, $t = 1.287$ and $p = 0.198$. This result is theoretically useful because it shows that not all marketplace cues depend on relational attachment in the same way. Scarcity may operate more through urgency, perceived loss and time pressure than through host attachment. These forces can become salient even when parasocial attachment is weaker, especially in live shopping settings where limited availability is visible to the audience in real time.

Self-esteem weakened the associations between all three marketplace cues and impulse buying tendency. The interaction was negative for perceived value, with $\beta = -0.10$, $t = 2.12$ and $p = 0.034$, serendipity, with $\beta = -0.08$, $t = 2.02$ and $p = 0.043$, and perceived scarcity, with $\beta = -0.07$, $t = 2.00$ and $p = 0.046$. This pattern is consistent with prior work linking lower self-esteem to compensatory and impulsive consumption (Dhandra, 2020; Challet-Bouju et al., 2020; Zhang, 2023). In live shopping, higher self-esteem may provide psychological distance from value claims, discovery appeals and scarcity pressure. Lower self-esteem, by contrast, may be associated with greater responsiveness to cues that appear to offer affirmation, gratification or reduced risk of exclusion.

The interaction coefficients are modest in magnitude, which is unsurprising in a large survey model involving psychological and relational conditions. Their value lies less in the size of any single coefficient than in the pattern they reveal. Perceived value, serendipity and scarcity show the strongest empirical associations with impulse buying tendency. Parasocial relationships and self-esteem qualify these associations by indicating when cue susceptibility becomes stronger or weaker. The study therefore contributes by showing that marketplace cues in live shopping are not associated with impulse buying tendency in a uniform way across consumers. Relational attachment and self-evaluation help explain variation in consumer susceptibility while preserving the main empirical role of value, discovery and scarcity.

5. Implications

The study advances research on impulse buying in live shopping by showing that marketplace cues are associated with impulse buying tendency. Perceived value, serendipity and scarcity show the strongest associations with impulse buying tendency. Parasocial relationships and self-esteem qualify these associations by indicating when consumers appear more or less susceptible to such cues. The contribution is therefore not that value, discovery and scarcity matter in isolation. It is that their association with consumer susceptibility varies across relational attachment and self-evaluation in a socially interactive shopping format.

5.1 Theoretical Implications

First, the study contributes to impulse buying research by connecting three elements that are often examined in separate streams: marketplace cues, parasocial attachment and self-esteem. Much prior work explains impulse buying through stimulus response reasoning, including the Stimulus-Organism-Response framework, where environmental cues are associated with internal states and consumer responses (Chan et al., 2017; Mehrabian and Russell, 1974; Redine et al., 2023). The present study adds an interpretive layer by showing that the same classes of cue are not associated with impulse buying tendency in a uniform way across consumers. In live shopping, cues are embedded in a social scene shaped by hosts, audiences and platform interaction. Their association with consumer susceptibility is therefore conditioned by relational and dispositional factors.

Second, the findings reposition parasocial relationships in impulse buying research. Prior studies have shown that parasocial relationships with influencers are associated with trust, engagement and purchase intention (Farivar et al., 2021; Xiang et al., 2016). The present results add to this literature by showing that parasocial attachment qualifies the associations between two marketplace cues and impulse buying tendency. The associations involving perceived value and serendipity were stronger among consumers with higher parasocial relationships. This pattern indicates that live shopping hosts do more than transmit product information. They provide a relational context in which value claims and unexpected discoveries may become more persuasive, familiar and socially affirmed.

Third, the findings refine the role of self-esteem in impulsive consumption. Prior research has linked lower self-esteem to greater susceptibility to compensatory and impulsive consumption (Dhandra, 2020; Reitz et al., 2020; Zhang, 2023). The present study shows that self-esteem also qualifies the association between marketplace cues and impulse buying tendency. Higher self-esteem weakened the associations involving perceived

value, serendipity and scarcity. This pattern positions self-esteem as a dispositional resource associated with psychological distance from value claims, discovery appeals and scarcity pressure in socially charged commercial settings.

Fourth, the study offers an Identity-Based Motivation reading of live shopping susceptibility. The results do not require the conclusion that consumers consciously translated every cue into identity content. They are consistent with the idea that cues encountered in live shopping may gain salience through taste, aspiration, belonging and social positioning. This reading helps explain why similar live shopping environments are associated with different levels of impulse buying tendency among consumers. It also preserves the contribution of established accounts based on arousal, urgency, self-regulation and goal conflict.

5.2 Practical Implications

For firms, the results imply that live shopping effectiveness should not depend on escalating promotional pressure. Value communication should be credible, specific and easy to evaluate through product demonstrations, live trials and question-and-answer exchanges. These practices can reduce uncertainty and make value claims more persuasive without relying on excessive urgency.

The results also point to the commercial value of discovery. Serendipity showed the strongest association with impulse buying tendency, which indicates that live shopping formats should preserve opportunities for surprise, novelty and exploration. Hosts can introduce unexpected product combinations, thematic product reveals and audience-guided discovery moments. Such practices can sustain engagement while keeping the shopping experience less mechanical and less reliant on price promotion.

Scarcity requires a different managerial approach. Scarcity cues were associated with impulse buying tendency, but the interaction with parasocial relationships was not statistically supported. This pattern indicates that scarcity may operate through urgency and perceived loss more than through host attachment. Managers should therefore use scarcity claims with restraint. Countdown timers, limited stock messages and exclusive offers should correspond to real constraints. Artificial urgency may generate short-term conversion, but it can erode trust and weaken relational engagement over time (Biraglia et al., 2021).

The findings concerning parasocial relationships have implications for influencer selection. Stronger parasocial relationships intensified the associations involving value and serendipity. Firms should therefore select hosts whose audience relationships are built on credibility, consistency and perceived fit with the product category. Transparent sponsorship disclosure, stable host brand alignment and responsive communication can strengthen relational trust while reducing perceptions of manipulative persuasion.

The findings concerning self-esteem also point to a need for consumer-centred design in live shopping. Consumers differ in their responsiveness to cues that appear to offer affirmation, gratification or reduced risk of exclusion. Communication strategies should therefore emphasise informed choice, product fit and consumer autonomy. Appeals that rely on insecurity, social comparison or fear of exclusion may create short-term sales but pose risks for consumer welfare.

At the policy level, the study supports stronger digital consumer literacy in influencer commerce. Platforms can provide visible sponsorship disclosures, budget prompts,

purchase pause options and information on stock claims. Regulators and educational institutions can help young consumers recognise how urgency, relational attachment and social visibility may affect purchasing decisions in live shopping. The aim is not to suppress live commerce, but to support a healthier balance between persuasion, participation and consumer autonomy.

5.3 Limitations and Directions for Future Research

Several scope conditions should guide the interpretation of the findings. First, the study used a cross-sectional survey design. The analysis identifies associations among marketplace cues, parasocial relationships, self-esteem and impulse buying tendency, but it cannot establish temporal ordering. Consumers with stronger impulse buying tendencies may, for instance, be more likely to develop parasocial attachment to live shopping hosts. Longitudinal research could examine how parasocial relationships, marketplace cue perceptions and impulse buying tendency evolve over time.

Second, the study measured impulse buying tendency, not purchase conduct during a specific live shopping episode. The Verplanken and Herabadi (2001) scale captures a dispositional inclination towards spontaneous and less deliberative buying. Asking respondents to answer with recent live shopping experiences in mind anchors the measure in the empirical setting, but preserves its status as a tendency measure. Future studies could complement this approach with episode-based purchase intentions, simulated purchase tasks, behavioural traces, clickstream data or transaction records.

Third, the study used a single-source survey design. Procedural safeguards were implemented, including anonymous participation, separation of incentive registration from survey responses and established measurement scales. Even so, common method effects cannot be fully ruled out. Full collinearity VIF values offer a diagnostic for multicollinearity and possible method concerns, but they do not constitute a test of common method variance. The design did not include a marker variable, a common latent method factor, temporal separation between predictor and criterion measures or data from several sources. The R squared value for impulse buying tendency and the interaction terms should therefore be interpreted as survey-based predictive associations. Future research could strengthen inference through longitudinal designs, experiments, behavioural data or data from several sources.

Fourth, Identity-Based Motivation was used as a conceptual lens, not as a measured psychological mechanism. The survey did not include items assessing whether respondents experienced products or marketplace cues as self-relevant, identity-congruent or intended for someone like them. The findings therefore support a restrained identity-oriented interpretation of conditional associations. Future research could measure self-relevance, identity congruence and perceived fit between products, influencers and consumer identity.

Fifth, the empirical setting was limited to Portuguese Generation Z consumers recruited through live shopping hosts with established digital audiences. This context is appropriate for examining digitally active young consumers, but it may overrepresent individuals who are already engaged with influencer-led commerce. Cross-national and cross-generational studies could assess whether similar associations emerge across markets with different levels of live shopping maturity, regulatory conditions and influencer cultures.

Future research could also extend the model by incorporating perceived authenticity, social comparison orientation, susceptibility to interpersonal influence, trust in the host

and perceived influencer credibility. These factors may refine understanding of why consumers differ in their responsiveness to marketplace cues in live shopping. Research on audience visibility, peer interaction and social contagion could also deepen understanding of how impulse buying tendency develops in socially interactive commerce environments.

References

Akram, U., Ansari, A.R. and Yan, C. (2023) 'Cosmetics makers have always sold "hope in a jar"! Understanding the cosmetics purchase intention in the Chinese mobile commerce environment', *Journal of Retailing and Consumer Services*, 73. Available at: https://doi.org/10.1016/j.jretconser.2023.103337

ANACOM (2024) *E-commerce in Portugal and the European Union in 2023 - residential and business segment*. Available at: https://www.anacom.pt/render.jsp?contentId=1773915 (Accessed: 26 February 2024).

Aw, E.C.X., Tan, G.W.H., Chuah, S.H.W., Ooi, K.B. and Hajli, N. (2022) 'Be my friend! Cultivating parasocial relationships with social media influencers: findings from PLS-SEM and fsQCA', *Information Technology & People*, 36(1), pp. 66-94. Available at: https://doi.org/10.1108/ITP-07-2021-0548

Bao, Z. and Zhu, Y. (2023) 'Understanding customers' stickiness of live streaming commerce platforms: an empirical study based on modified e-commerce system success model', *Asia Pacific Journal of Marketing and Logistics*, 35(3), pp. 775-793. Available at: https://doi.org/10.1108/APJML-09-2021-0707

Bashar, A., Singh, S. and Pathak, V.K. (2023) 'The influence of culture on impulse buying behavior: a systematic literature review', *BBR: Brazilian Business Review*, 20(4), pp. 465-484. https://doi.org/10.15728/bbr.2022.1221.en

Biraglia, A., Usrey, B. and Ulqinaku, A. (2021) 'The downside of scarcity: Scarcity appeals can trigger consumer anger and brand switching intentions', *Psychology & Marketing*, 38(8), pp. 1314-1322. Available at: https://doi.org/10.1002/mar.21489

Brehm, J.W. (1989). *Psychological Reactance: Theory and Applications*. ACR North American Advances.

Busch, C. and Grimes, M. (2023) 'Eureka!: Improving the effects of artificial intelligence on serendipitous innovation', *Academy of Management Proceedings*, 2023(1), p. 11339. Available at: https://doi.org/10.5465/AMPROC.2023.11339abstract

Challet-Bouju, G., Mariez, J., Perrot, B., Grall-Bronnec, M. and Chauchard, E. (2020) 'A typology of buyers grounded in psychological risk factors for compulsive buying', *Frontiers in Psychiatry*, 11, p. 277. Available at: https://doi.org/10.3389/fpsyt.2020.00277

Chan, T.K., Cheung, C.M. and Lee, Z.W. (2017) 'The state of online impulse-buying research: A literature analysis', *Information & Management*, 54(2), pp. 204-217. Available at: https://doi.org/10.1016/j.im.2016.06.001

Chen, J.V., Ha, Q.A. and Vu, M.T. (2023) 'The influences of virtual reality shopping characteristics on consumers' impulse buying behavior', *International Journal of Human-Computer Interaction*, 39(17), pp. 3473-3491. https://doi.org/10.1080/10447318.2022.2098566

Chen, W.K., Chen, C.W. and Lin, Y.C. (2023) 'Understanding the influence of impulse buying toward consumers' post-purchase dissonance and return intention', *Journal of Ambient Intelligence and Humanized Computing*, pp. 1-14. Available at: https://doi.org/10.1007/s12652-020-02333-z

Chung, S. and Cho, H. (2017) 'Fostering parasocial relationships with celebrities on social media', *Psychology & Marketing*, 34(4), pp. 481-495. Available at: https://doi.org/10.1002/mar.21001

Dhandra, T.K. (2020) 'Does self-esteem matter?', *Journal of Retailing and Consumer Services*, 55, p. 102135. Available at: https://doi.org/10.1016/j.jretconser.2020.102135

Djafarova, E. and Bowes, T. (2021) 'Instagram made me buy it', *Journal of Retailing and Consumer Services*, 59, p. 102345. Available at: https://doi.org/10.1016/j.jretconser.2020.102345

Dodds, W.B., Monroe, K.B. and Grewal, D. (1991) 'Effects of price, brand, and store information on buyers' product evaluations', *Journal of Marketing Research*, 28(3), pp. 307-319. Available at: https://doi.org/10.2307/3172866

Ekayani, N.P.A., Sukawati, T.G.R. and Yasa, N.N.K. (2024) 'The role of brand image mediates the effect of product quality on repurchase intention', *International Journal of Asian Business and Management*, 3(4), pp. 525-540. Available at: https://doi.org/10.55927/ijabm.v3i4.10282

Escalas, J.E. and Bettman, J.R. (2005) 'Self-construal, reference groups, and brand meaning', *Journal of Consumer Research*, 32(3), pp. 378-389. Available at: https://doi.org/10.1086/497549

Farivar, S., Wang, F. and Yuan, Y. (2021) 'Opinion leadership vs. para-social relationship', *Journal of Retailing and Consumer Services*, 59, p. 102371. Available at: https://doi.org/10.1016/j.jretconser.2020.102371

Feng, Z. et al. (2024) 'Impulse buying behavior during livestreaming', *Heliyon*, 10(7). Available at: https://doi.org/10.1016/j.heliyon.2024.e28347

Fornell, C. and Larcker, D.F. (1981) 'Evaluating structural equation models', *Journal of Marketing Research*, 18(1), pp. 39-50. Available at: https://doi.org/10.2307/3151312

Goel, P. et al. (2022) 'A moderated mediation model for e-impulse buying tendency', *Journal of Business Research*, 142, pp. 1-16. Available at: https://doi.org/10.1016/j.jbusres.2021.12.041

Gunawan, N.P. and Iskandar, I.B.P. (2020) 'Analyzing the impact of fashion influencer', *KnE Social Sciences*, pp. 350-363. Available at: https://doi.org/10.18502/kss.v4i6.6611

Gupta, S. and Gentry, J.W. (2016) 'The behavioral responses to perceived scarcity', *The International Review of Retail, Distribution and Consumer Research*, 26(3), pp. 260-271. Available at: https://doi.org/10.1080/09593969.2016.1147476

Hair, J., Risher, J., Sarstedt, M. and Ringle, C. (2019) 'When to use and how to report the results of PLS-SEM', *European Business Review*, 31(1), pp. 2-24. Available at: https://doi.org/10.1108/EBR-11-2018-0203

Hayes, A. F. (2022). *Introduction to mediation, moderation, and conditional process analysis: A Regression-based Approach*. Guilford Publications.

Henseler, J., Ringle, C.M. and Sarstedt, M. (2015) 'A new criterion for assessing discriminant validity', *Journal of the Academy of Marketing Science*, 43(1), pp. 115-135. Available at: https://doi.org/10.1007/s11747-014-0403-8

Higgins, E.T. (1997) 'Beyond pleasure and pain', *American Psychologist*, 52(12), p. 1280. Available at: https://doi.org/10.1037/0003-066X.52.12.1280

Hoffner, C.A. and Bond, B.J. (2022) 'Parasocial relationships, social media, & well-being', *Current Opinion in Psychology*, 45. Available at: https://doi.org/10.1016/j.copsyc.2022.101306

Hu, L., Min, Q., Han, S. and Liu, Z. (2020) 'Understanding followers' stickiness to digital influencers', *International Journal of Information Management*, 54, p. 102169. Available at: https://doi.org/10.1016/j.ijinfomgt.2020.102169

Jin, S.V. and Ryu, E. (2020) 'I'll buy what she's wearing', *Journal of Retailing and Consumer Services*, 55, p. 102121. Available at: https://doi.org/10.1016/j.jretconser.2020.102121

Kim, A., Affonso, F.M., Laran, J. and Durante, K.M. (2021) 'Serendipity', *Journal of Marketing*, 85(4), pp. 141-157. Available at: https://doi.org/10.1177/00222429211000344

Kim, H.J. and Chan-Olmsted, S. (2022) 'Influencer marketing and social commerce', *Journal of Interactive Advertising*, 22(3), pp. 249-268. Available at: https://doi.org/10.1080/15252019.2022.2111243

Kimiagari, S. and Malafe, N.S.A. (2021) 'The role of cognitive and affective responses', *Journal of Retailing and Consumer Services*, 61, p. 102567. Available at: https://doi.org/10.1016/j.jretconser.2021.102567

Kock, N. (2015) 'Common method bias in PLS-SEM', *International Journal of e-Collaboration*, 11(4), pp. 1-10. Available at: https://doi.org/10.4018/ijec.2015100101

Lee, J.E. and Watkins, B. (2016) 'YouTube vloggers' influence on consumer luxury brand perceptions', *Journal of Business Research*, 69(12), pp. 5753-5760. Available at: https://doi.org/10.1016/j.jbusres.2016.04.171

Leroi-Werelds, S. (2019) 'An update on customer value', *Journal of Service Management*, 30(5), pp. 650-680. Available at: https://doi.org/10.1108/JOSM-03-2019-0074

Li, L., Kang, K., Zhao, A. and Feng, Y. (2022) 'The impact of social presence and facilitation factors', *Information Technology & People*. Available at: https://doi.org/10.1108/ITP-03-2021-0203

Liu, Y., Qin, C., Ma, X. and Liang, H. (2022) 'Serendipity in human information behavior', *Journal of Documentation*, 78(2), pp. 435-462. Available at: https://doi.org/10.1108/JD-02-2021-0029

Malik, A., Gulzar, A., Khan, M.A. and Siddique, N. (2024) 'Mapping global research on Generation Z', *Global Knowledge, Memory and Communication*. Available at: https://doi.org/10.1108/GKMC-07-2024-0428

Mardon, R., Cocker, H., & Daunt, K. (2023). When parasocial relationships turn sour: social media influencers, eroded and exploitative intimacies, and anti-fan communities. *Journal of Marketing Management, 39*(11-12), 1132-1162. https://doi.org/10.1080/0267257X.2022.2149609

McCay-Peet, L. and Toms, E. (2011) 'Measuring the dimensions of serendipity in digital environments', *Information Research*, 16(3), p. n3.

Mehrabian, A. and Russell, J.A. (1974) *An approach to environmental psychology*. MIT Press.

Ming, J., Jianqiu, Z., Bilal, M., Akram, U. and Fan, M. (2021) 'How social presence influences impulse buying behavior in live streaming commerce? The role of S-O-R theory', *International Journal of Web Information Systems*, 17(4), pp. 300-320. Available at: https://doi.org/10.1108/IJWIS-02-2021-0012

Musadik, S.H.S.A. (2021) 'A review: Scarcity and impulse buying behavior amid COVID-19 pandemic', *International Journal of Business and Economy*, 3(4), pp. 180-189.

Omar, N.A., Nazri, M.A., Ali, M.H. and Alam, S.S. (2021) 'The panic buying behavior of consumers during the COVID-19 pandemic', *Journal of Retailing and Consumer Services*, 62, p. 102600. Available at: https://doi.org/10.1016/j.jretconser.2021.102600

Oyserman, D. (2009) 'Identity-based motivation: Implications for action-readiness, procedural-readiness, and consumer behavior', *Journal of Consumer Psychology*, 19(3), pp. 250-260. Available at: https://doi.org/10.1016/j.jcps.2009.05.008

Oyserman, D. (2015) 'Identity-based motivation', in Scott, R.A. and Kosslyn, S.M. (eds.) *Emerging Trends in the Social and Behavioral Sciences*. Hoboken, NJ: Wiley. Available at: https://doi.org/10.1002/9781118900772.etrds0171

Pacheco, D. C., Moniz, A. I. D. S. A., Caldeira, S. N. and Silva, O. D. L. (2026). The price of belonging: impulse buying in live shopping among low-income youth. *Qualitative Market Research: An International Journal*, *29*(2), pp. 286-312. https://doi.org/10.1108/QMR-10-2025-0244

Pacheco, D.C., Moniz, A.I.D.S.A., Silva, O.D.L., Caldeira, S.N. and Bigné, E. (2025) 'Impulsivity in live shopping', *International Journal of Consumer Studies*, 49(5), p. e70113. Available at: https://doi.org/10.1111/ijcs.70113

Parameswari, J., Yoganandan, G. and Vasan, M. (2024) 'Mediating effect of self-esteem', *Entertainment Computing*, 50, p. 100650. Available at: https://doi.org/10.1016/j.entcom.2024.100650

Prawira, N.A. and Sihombing, S.O. (2021) 'Predicting the relationship between scarcity and serendipity information', *ASEAN Marketing Journal*, 13(1), p. 5. Available at: https://doi.org/10.21002/amj.v13i1.13216

Qiao, Y., Yin, X. and Xing, G. (2022) 'Impact of perceived product value', *Frontiers in Psychology*, 13, p. 931064. Available at: https://doi.org/10.3389/fpsyg.2022.931064

Redine, A., Deshpande, S., Jebarajakirthy, C. and Surachartkumtonkun, J. (2023) 'Impulse buying', *International Journal of Consumer Studies*, 47(1), pp. 3-41. Available at: https://doi.org/10.1111/ijcs.12862

Reitz, A.K., Shrout, P.E., Denissen, J.J., Dufner, M. and Bolger, N. (2020) 'Self-esteem change', *Journal of Personality*, 88(4), pp. 689-702. Available at: https://doi.org/10.1111/jopy.12519

Ringle, C.M., Wende, S. and Becker, J. (2024) *SmartPLS 4*. Available at: https://www.smartpls.com

Rook, D.W. and Hoch, S.J. (1985) 'Consuming impulses', *Advances in Consumer Research*.

Rosenberg, M. (1979) *Conceiving the self*. Basic Books.

Rosenberg, M., Schooler, C., Schoenbach, C. and Rosenberg, F. (1995) 'Global self-esteem and specific self-esteem', *American Sociological Review*, pp. 141-156. Available at: https://doi.org/10.2307/2096350

Sheng, J., Kostyk, A., Chen, J. and Wiggins, J. (2025) 'From parasocial interaction to parasocial relationship: A review and research agenda', *International Journal of Consumer Studies*, 49, e70038. Available at: https://doi.org/10.1111/ijcs.70038

Shi, X., Li, F. and Chumnumpan, P. (2020) 'The use of product scarcity in marketing', *European Journal of Marketing*, 54(2), pp. 380-418. Available at: https://doi.org/10.1108/EJM-04-2018-0285

Song, M., Choi, S. and Moon, J. (2021) 'Limited time or limited quantity?', *Journal of Hospitality and Tourism Management*, 47, pp. 167-175. Available at: https://doi.org/10.1016/j.jhtm.2021.03.012

Soper, D.S. (2025) *A-priori sample size calculator for structural equation models* [Software]. Available at: https://www.danielsoper.com/statcalc

Sun, Y., Shao, X., Li, X., Guo, Y. and Nie, K. (2019) 'How live streaming influences purchase intentions', *Electronic Commerce Research and Applications*, 37, p. 100886. Available at: https://doi.org/10.1016/j.elerap.2019.100886

Sweeney, J.C. and Soutar, G.N. (2001) 'Consumer perceived value', *Journal of Retailing*, 77(2), pp. 203-220. Available at: https://doi.org/10.1016/S0022-4359(01)00041-0

Tukachinsky, R. (2011) 'Para-romantic love and para-friendships: Development and assessment of a multiple-parasocial relationships scale', *American Journal of Media Psychology*, 3(1/2), pp. 73-94.

Verplanken, B. and Herabadi, A. (2001) 'Individual differences in impulse buying tendency', *European Journal of Personality*, 15, pp. S71-S83. Available at: https://doi.org/10.1002/per.423

Walsh, G., Shiu, E. and Hassan, L.M. (2014) 'Replicating, validating, and reducing the length of the consumer perceived value scale', *Journal of Business Research*, 67(3), pp. 260-267. Available at: https://doi.org/10.1016/j.jbusres.2013.05.012

Westland, J.C. (2010) 'Lower bounds on sample size in structural equation modeling', *Electronic Commerce Research and Applications*, 9(6), pp. 476-487. Available at: https://doi.org/10.1016/j.elerap.2010.07.003

Wiener, H.J., Wiener, J. and Arnold, T. (2022) 'Enhancing perceived product value', *European Journal of Marketing*, 56(2), pp. 375-399. Available at: https://doi.org/10.1108/EJM-07-2020-0538

Wu, W.Y., Lu, H.Y., Wu, Y.Y. and Fu, C.S. (2011) 'The effects of product scarcity', *International Journal of Consumer Studies*, 36(3), pp. 263-274. Available at: https://doi.org/10.1111/j.1470-6431.2011.01000.x

Xiang, L., Zheng, X., Lee, M.K. and Zhao, D. (2016) 'Exploring consumers' impulse buying behavior on social commerce platform', *International Journal of Information Management*, 36(3), pp. 333-347. Available at: https://doi.org/10.1016/j.ijinfomgt.2015.11.002

Yaqub, O. (2018) 'Serendipity', *Research Policy*, 47(1), pp. 169-179. Available at: https://doi.org/10.1016/j.respol.2017.10.007

Yi, Q., Khan, J., Su, Y., Tong, J. and Zhao, S. (2023) 'Impulse buying tendency in live-stream commerce', *Journal of Retailing and Consumer Services*, 75, p. 103534. Available at: https://doi.org/10.1016/j.jretconser.2023.103534

Zhang, N. (2023) 'Product presentation in the live-streaming context', *Frontiers in Psychology*, pp. 1-12. Available at: https://doi.org/10.3389/fpsyg.2023.1124675

Zhang, Y. and Hawk, S.T. (2022) 'I buy stability in a buying world', *Personality and Individual Differences*, 184, p. 111184. Available at: https://doi.org/10.1016/j.paid.2021.111184

Zuliarni, S., Kartikasari, D., Hendrawan, B. and Siregar, S.S.W. (2023) 'The impact of buying intention of global fashion on local substitute', *Heliyon*, 9(11). Available at: https://doi.org/10.1016/j.heliyon.2023.e22160

Appendix A. Measurement Scales and Factor Loadings

Table A1. Measurement scales and standardised factor loadings

Original scale	Construct	Items		Loadings	CA	CR	AVE
Verplanken and Herabadi (2001)	Impulse Buying Tendency	**Instruction:** The following items describe different ways people approach buying decisions. Although these statements refer to your general buying tendencies, please answer them with your recent live shopping experiences in mind.					
		IBT1.	I usually think carefully before I buy something. (R)	0.72	0.91	0.94	0.65
		IBT2.	If I buy something, I usually do that spontaneously.	0.84			
		IBT3.	I like to compare different brands before I buy one. (R)	0.70			
		IBT4.	I am used to buying things 'on the spot'.	0.86			
		IBT5.	I often buy things without thinking.	0.88			
		IBT6.	It is a struggle to leave nice things I see in a shop.	0.82			
		IBT7.	I sometimes cannot suppress the feeling of wanting to buy something.	0.85			
		IBT8.	I sometimes feel guilty after having bought something.	0.76			
		IBT9.	I always see something nice whenever I pass by shops.	0.78			
		IBT10.	I am a bit reckless in buying things.	0.83			
Walsh et al. (2014)	Perceived Value	**Instruction:** The following items assess your perceptions of the products presented during live shopping sessions by the influencer. Please answer based on your recent live shopping experiences.			0.88	0.93	0.66
		PV1.	The products had consistent quality.	0.82			
		PV2.	The products had poor workmanship (R).	0.71			
		PV3.	The products were something I enjoyed.	0.86			
		PV4.	The products made me want to use them.	0.88			
		PV5.	The products gave me pleasure.	0.87			
		PV6.	The products were reasonably priced.	0.80			
		PV7.	The products offered value for money.	0.84			
		PV8.	The products would help me to feel acceptable.	0.76			
		PV9.	The products would give its owner social approval.	0.74			
McCay-Peet and Toms (2011)	Serendipity	**Instruction:** The following items describe experiences that may occur while watching live shopping sessions, including encountering products unexpectedly during live presentations. Please answer considering your recent live shopping experiences.			0.93	0.88	0.70
		S1.	I came across unexpected and interesting products during the streams.	0.83			
		S2.	The streams encouraged me to stop what I was doing and get to know the products being presented.	0.87			
		S3.	I explored many products that I do not regularly examine.	0.82			
Wu et al. (2011)	Perceived Scarcity	**Instruction:** The following items assess your perceptions of product availability during live shopping sessions. Please indicate the extent to which you agree based on your recent live shopping experiences.			0.89	0.88	0.66
		PSc1.	I thought that the supply of the products was small.	0.840			
		PSc2.	I thought that the products were selling out soon.	0.865			
		PSc3.	I thought that the limited supply would cause a lot of people to buy them.	0.862			
Aw et al. (2022)	Parasocial relationships	**Instruction:** The following items assess your perceived relationship with the live shopping influencer (i.e., the person presenting products during live sessions). Please indicate the extent to which you agree with each statement based on your recent live shopping experiences.			0.90	0.92	0.66
		PR1.	The live shopping vendor makes me feel comfortable, as if I were with a friend.	0.85			
		PR2.	I would like to have a friendly chat with the live shopping vendor.	0.87			
		PR3.	I think I understand the live shopping vendor quite well.	0.83			

		PR4.	I can feel the live shopping vendor's emotions in certain situations.	0.84			
		PR5.	The live shopping vendor reminds me of myself.	0.78			
		PR6.	I can identify with the live shopping vendor.	0.88			
Rosenberg (1979)	Self-esteem	**Instruction:** The following items assess your general self-evaluation. Please indicate the extent to which you agree with each statement.					
		SE1.	On the whole, I am satisfied with myself.	0.86	0.91	0.92	0.63
		SE2.	At times I think I am no good at all. (R)	0.73			
		SE3.	I feel that I have a number of good qualities.	0.88			
		SE4.	I am able to do things as well as most other people.	0.84			
		SE5.	I feel that I'm a person of worth.	0.87			
		SE6.	I take a positive attitude toward myself.	0.89			

Participants were instructed to respond based on their recent live shopping experiences, allowing contextual cues to be interpreted in relation to their personal perceptions.

Note. (R) = reverse-coded items (recoded prior to analysis).

All loadings are standardised and significant at $p < .001$.

Impulse Buying Tendency was measured using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree); Perceived Value was assessed through a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree); Serendipity was measured using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree); Perceived Scarcity was evaluated using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree); Parasocial Relationships were measured using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree); and Self-Esteem was assessed using a 4-point Likert scale (1 = strongly disagree, 4 = strongly agree).